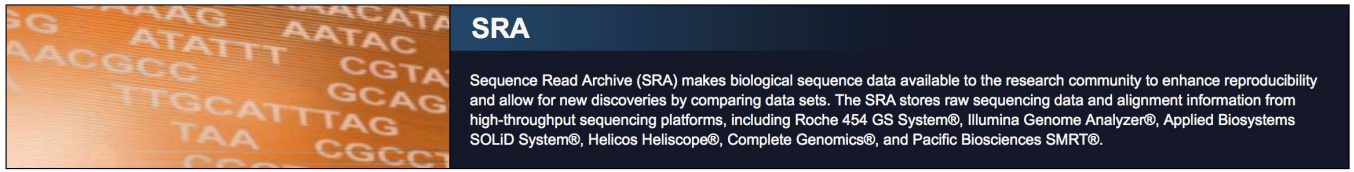
Databases and Genome Browsers

- Databases
- Genome browsers
  - UCSC
  - IGV

CM 515

Data Repositories and Deposition of Data

SRA: Sequence Read Archive – raw NextGen sequencing data (fastq files)



GEO: Gene Expression Omnibus – minimally processed genomics datasets (fastq files + tables). Most journals require that you submit your data to GEO. GEO deposits raw data to SRA.

GenBank: A repository of gene and genome sequences (fasta files).

Exercise 1: Retrieving data from GEO and GenBank

Genome Information Databases

Genome information is available for many common model organisms in web accessible databases.

Model Organisms

Database	Organism	Organism (Genus and species)
BeeBase	Honey Bee	Apis mellifera
BeetleBase	Red Flour Beetle	Tribolium castaneum
DictyBase	Soil Amoeba	Dictyostelium discoideum
EcoCyc	Bacteria	Escherichia coli
Gramene	Grasses	
MGI	Mouse	Mus musculus
OMIM	Human	Homo sapiens
SGD	Budding Yeast	Saccharomyces cerevisiae
S. pombe GeneDB	Fission Yeast	Schizosaccharomyces pombe
TAIR	Mouse-ear cress	Arabidopsis thaliana
VectorBase	Mosquitos and Deer Tick	Anopheles gambiae, Aedes aegypti, Culex pipiens quinquefasciatus, Ixodes scapularis
WormBase	Round worms	Caenorhabditis
ZFIN	Zebrafish	Danio rerio

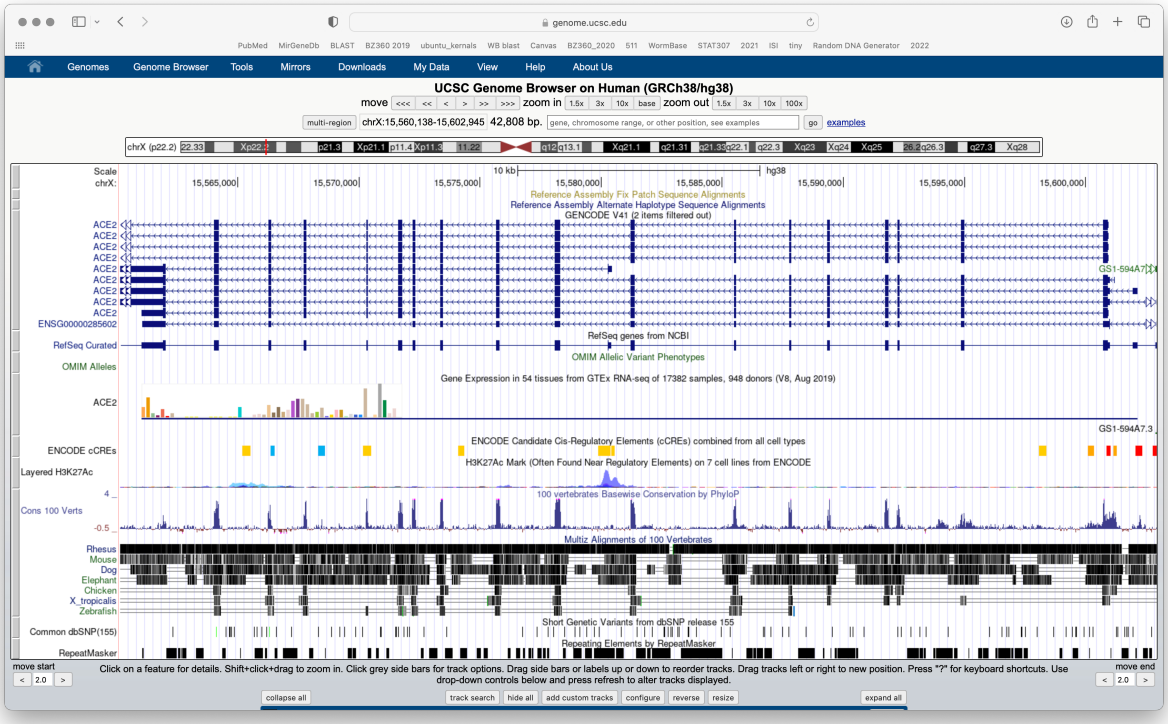
Many databases are part of the Generic Model Organism Database project.

Many are also part of the Alliance of Genome Resources consortium

Some other useful sources of genome information include UCSC and Ensembl:

Genome Browsers and Databases

Genome browsers are graphical interfaces for displaying information from a database on a genome representation. Features often include genes and expression data. UCSC is one example that we'll look at today.



We'll explore two databases today:

1. UCSC
2. Ensemble

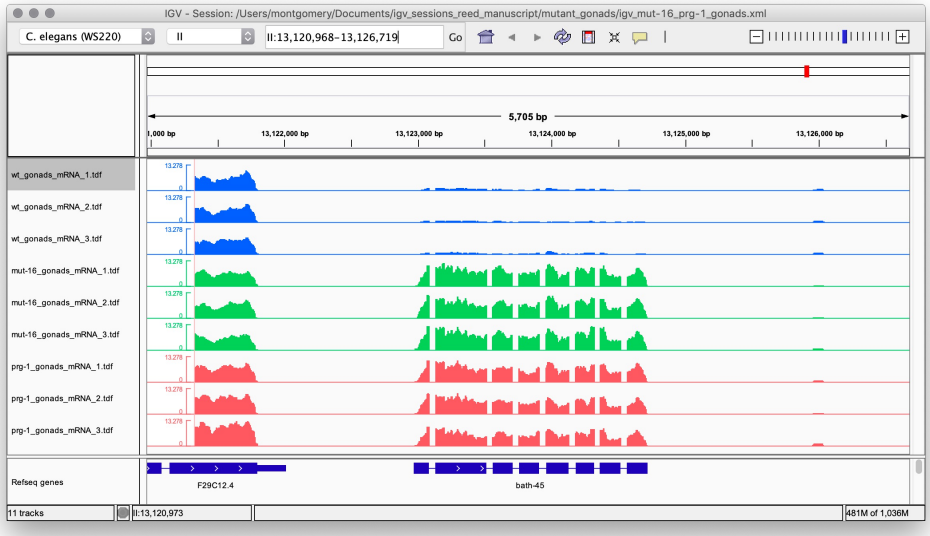
UCSC Genome Bioinformatics

Ensembl

Exercise 2: Exploring organismal databases and web-based genome browsers

Integrative Genomics Viewer (IGV)

A desktop browser for viewing genomics data in the context of a genome.



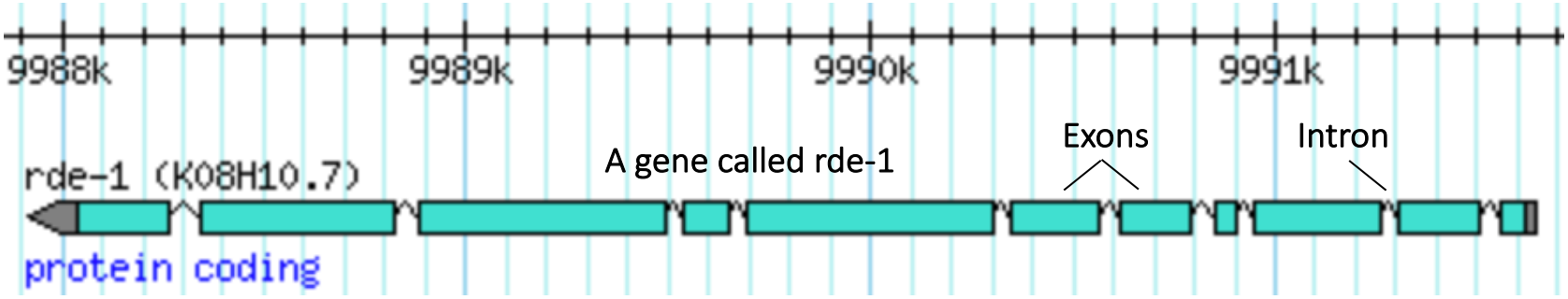
The histograms in this IGV screenshot show coverage profiles for several libraries, with each line in the histogram representing the read depth at that genomic position. The tracks are colored by sample group.

Exercise 3: Data visualization in IGV

GFF3/GTF Feature Format

Positional/annotation information about sequence features, such as genes, is often stored as a gff3 or gtf formatted file.

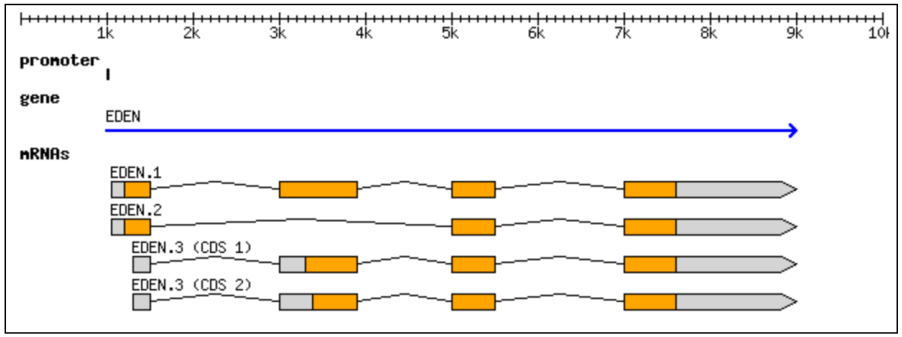
Position on chromosome V relative to start of chromosome




```
#Typical gff file format for gene annotations
CHROMOSOME    SOURCE    FEATURE    START    END      SCORE    STRAND    FRAME    NOTE
V             WormBase  gene       9987196  9991642  .        -          .       Name=rde-1
```

Generic Feature Format (gff3): 9 tab-delimited columns

GFF/GTF Format




```
ctg123 . exon    1300  1500  .  +  .  Parent=mRNA00003
ctg123 . exon    1050  1500  .  +  .  Parent=mRNA00001,mRNA00002
ctg123 . exon    3000  3902  .  +  .  Parent=mRNA00001,mRNA00003
ctg123 . exon    5000  5500  .  +  .  Parent=mRNA00001,mRNA00002,mRNA00003
ctg123 . exon    7000  9000  .  +  .  Parent=mRNA00001,mRNA00002,mRNA00003
ctg123 . CDS     1201  1500  .  +  0  ID=cds00001;Parent=mRNA00001
ctg123 . CDS     3000  3902  .  +  0  ID=cds00001;Parent=mRNA00001
ctg123 . CDS     5000  5500  .  +  0  ID=cds00001;Parent=mRNA00001
ctg123 . CDS     7000  7600  .  +  0  ID=cds00001;Parent=mRNA00001
ctg123 . CDS     1201  1500  .  +  0  ID=cds00002;Parent=mRNA00002
ctg123 . CDS     5000  5500  .  +  0  ID=cds00002;Parent=mRNA00002
ctg123 . CDS     7000  7600  .  +  0  ID=cds00002;Parent=mRNA00002
ctg123 . CDS     3301  3902  .  +  0  ID=cds00003;Parent=mRNA00003
ctg123 . CDS     5000  5500  .  +  1  ID=cds00003;Parent=mRNA00003
ctg123 . CDS     7000  7600  .  +  1  ID=cds00003;Parent=mRNA00003
ctg123 . CDS     3391  3902  .  +  0  ID=cds00004;Parent=mRNA00003
ctg123 . CDS     5000  5500  .  +  1  ID=cds00004;Parent=mRNA00003
ctg123 . CDS     7000  7600  .  +  1  ID=cds00004;Parent=mRNA00003
```

http://www.sequenceontology.org/gff3.shtml

Exercise 4: Parsing gff3 files with python